

Central Force

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In this chapter, we will apply Newton's laws to central forces. Any force will be a central force if it acts along the straight line connecting two bodies and its magnitude depends only on the distance r between them. If we consider one of the two bodies as the origin of our coordinate system, then in vector notation it can be written as-

$$\vec{F} = F\hat{r}$$

For such a general central force, we will determine the equation of motion. Next, using Newton's gravitational force $F(r) \propto \frac{1}{r^2}$, we will derive Kepler's laws.

The torque on an object due to a central force will be, $\vec{\tau} = \vec{r} \times F\hat{r} = 0$. Therefore, angular momentum will be conserved with respect to our chosen origin. Following Newton's third law, angular momentum will also be conserved with respect to any other center. If the mass of the rotating object is m , then its angular momentum is

$$\vec{l} = m\vec{r} \times \dot{\vec{r}}$$

That is, $\vec{l}$ is perpendicular to the plane containing the two vectors $\vec{r}$ and $\dot{\vec{r}}$. Since $\vec{l}$ is a constant vector, its magnitude and direction must remain the same. Again, if the direction of $\vec{l}$ is constant, the plane perpendicular to it can never change. Therefore, $\vec{r}$ and $\dot{\vec{r}}$ will always lie in this same plane. For this reason, any orbit under a central force is two-dimensional. Until Section 7.4, we will only work with a stationary central force source. That is, two objects will influence each other through a central force and one object will remain stationary. Furthermore, unless specified otherwise, we will always assume the stationary object is at the origin and work in a polar coordinate system.

1 Equation of Orbit under Central Force

In this section, we will attempt to determine the equation of motion for an object of mass m under the influence of any central force $\vec{F}$. In this section and throughout the central force topic, our main focus will be on conserved quantities: angular momentum and total energy. First, angular momentum

$$\vec{l} = m\vec{r} \times \dot{\vec{r}} = mrr_{\perp}\dot{\theta}$$

where $\dot{r}_{\perp}$ is the component of $\dot{\vec{r}}$ perpendicular to $\vec{r}$. We can write $\dot{r}_{\perp}$ in terms of angular velocity as $\dot{\theta} \times r$. Then,

$$\begin{aligned} l &= m\dot{\theta}r^2 \\ \Rightarrow \dot{\theta} &= \frac{l}{mr^2} \end{aligned} \tag{1}$$

We will use another variable instead of r , $u = \frac{1}{r}$. In that case, $\dot{\theta} = \frac{lu^2}{m}$. Now we can write,

$$\frac{d}{dt} = \frac{d}{d\theta} \frac{d\theta}{dt} = \frac{lu^2}{m} \frac{d}{d\theta}$$

Then,

$$\frac{d}{dt}r = \frac{d}{dt} \frac{1}{u} = \frac{lu^2}{m} \frac{d}{d\theta} \frac{1}{u} = -\frac{l}{m} \frac{du}{d\theta}$$

Again, to determine the second derivative-

$$\frac{d}{dt} \left(\frac{d}{dt}r \right) = \frac{-l}{m} \frac{d}{dt} \left(\frac{du}{d\theta} \right) = \frac{-l^2 u^2}{m^2} \frac{d^2 u}{d\theta^2} \quad (2)$$

To obtain the last equation, we have applied the chain rule again. Now, from Newton's second law along the radial direction, we get-

$$m\ddot{r} - m\dot{\theta}^2 r = F$$

Substituting $\ddot{r}$ and $\dot{\theta}$ from equation (1) and equation (2),

$$\begin{aligned} -\frac{l^2 u^2}{m} \frac{d^2 u}{d\theta^2} - \frac{l^2 u^3}{m} &= F \\ \Rightarrow \frac{d^2 u}{d\theta^2} + u &= -\frac{mF}{l^2 u^2} \end{aligned}$$

By substituting the value of F into this equation, a differential equation for $u(\theta)$ is obtained. Solving this equation makes it possible to determine $u(\theta)$ and from there $r(\theta)$.

For gravitational force, $\vec{F} = -GMmu^2\hat{r}$. Therefore, $F = -GMmu^2$. Substituting this,

$$\frac{d^2 u}{d\theta^2} + u = \frac{GMm^2}{l^2}$$

which is the differential equation for simple harmonic motion. Its solution is

$$u(\theta) = \frac{1}{r(\theta)} = A \cos(\theta + \phi) + \frac{GMm^2}{l^2} \quad (3)$$

$$r(\theta) = \frac{\frac{l^2}{GMm^2}}{1 + \frac{l^2 A}{GMm^2} \cos(\theta + \phi)}$$

The constants A and ϕ in the solution depend on the initial conditions of the system. Conventionally, the point closest to the orbit is taken as $\theta = 0$. That is why $\phi = 0$. Also, $\frac{l^2 A}{GMm^2}$ is called the eccentricity of the orbit. It is denoted by e , and the shape of the orbit depends entirely on it. Thus, the general equation of the orbit under gravitational force is,

$$r(\theta) = \frac{\frac{l^2}{GMm^2}}{1 + e \cos \theta} \quad (4)$$

The eccentricity e of the orbit depends on the total energy and angular momentum of the object. Another equation for it can be derived (you will derive this in Exercise 1),

$$e = \sqrt{1 + \frac{2El^2}{G^2 M^2 m^3}}$$

Note that since the closest point is taken as $\phi = 0^\circ$, e can never be negative.

For different values of e in equation (4), orbits of different shapes are obtained.

- $e \equiv 0$: For $e = 0$, the total energy E must be negative. In this case, according to equation (4), the orbit will be circular, and the radius of the circle is $r = \frac{l^2}{GMm^2}$.
- $0 < e < 1$: For $0 < e < 1$ as well, the total energy E must be negative. According to equation (4), the equation obtained in this case is the equation of an ellipse.
- $e \equiv 1$: In this case, $E = 0$ must hold. The resulting orbit is a parabola.
- $e > 1$: In this case, the value of E is greater than 0. Here, the orbit will be hyperbolic.

2 Conics

The four curves obtained for the different values of e discussed in the previous section are collectively called conic sections. In this part, we will learn a little more detail about these conic sections.

Circle

If $\vec{e} = \vec{0}$, we get a circle. This is the most general conic section. A circle has only one focus, which is located at its center. To completely specify a circle, knowing its radius and center is sufficient. The equation of a circle of radius R with respect to its focus (or center) is $r(\theta) = R$.

Ellipse

Figure 1 shows an ellipse. In the figure, AC is the major axis and BD is the minor axis of the ellipse. Their halves, OC and OD, are the semi-major (a) and semi-minor (b) axes, respectively. An ellipse has two foci (F and F'). They are located at equal distances from the geometric center of the ellipse and lie on the major axis. An interesting property of an ellipse is that for any point X on the ellipse, the sum ($F'X + XF$) always remains constant.

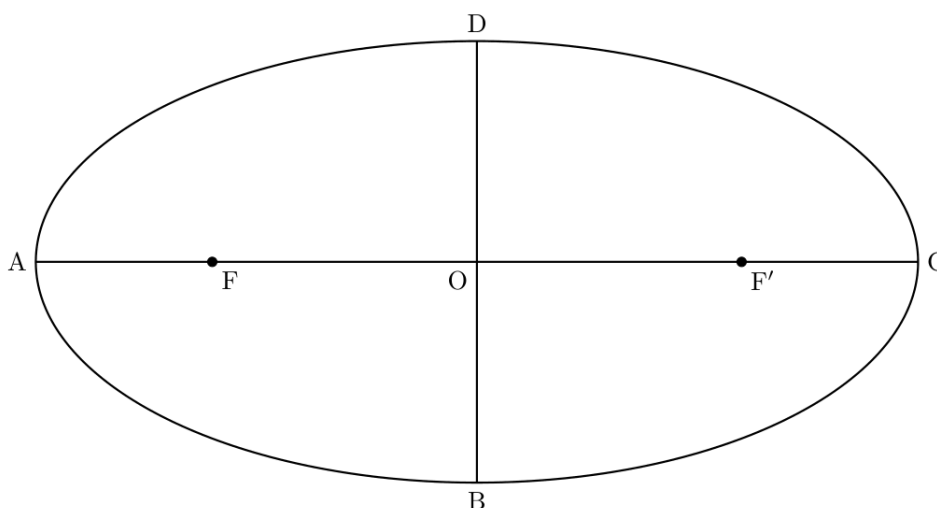


Figure 1: Ellipse

Its equation with respect to the center of the ellipse is

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$$

Its equation with respect to the focus (equation 4) is,

$$r = \frac{a(1 - e^2)}{1 + e \cos \theta} \quad (5)$$

Equation (5) gives the correct equation with respect to both foci. However, it must be kept in mind that we have already assumed the convention of $\theta = 0$. Accordingly, with respect to focus F , $\overline{FB}$ corresponds to $\theta = 0$. On the other hand, for focus F' , $\theta = 0$ is $\overline{F'A}$. After some geometry, you will be able to show that the distance of each focus from the center of the ellipse is $OF = OF' = ea$.

By now you might have noticed that if the e of an ellipse is very close to zero, it becomes a circle. Therefore, an ellipse has all the properties of a circle, but since a circle has both its foci at the same point, it basically has one focus (the center of the circle). Again, if e is very close to 1, it turns into a line segment. In that case, the semi-minor axis of the ellipse $b \rightarrow 0$ and the two foci lie at the two ends of the major axis. Note that if e of an ellipse approaches zero, a circle emerges, but if e approaches 1, a parabola does not emerge. The reason for this is that the equation for conic sections is not continuous at $e = 1$, but it is continuous at $e = 0$.

Parabola

For $e = 1$ we get a parabola. The general equation of a parabola is

$$y^2 = -4ax$$

and it has only one focus, which is located at $(-a, 0)$. Writing its equation in polar coordinates with respect to the focus yields:

$$r(\theta) = \frac{2a}{1 + \cos \theta}$$

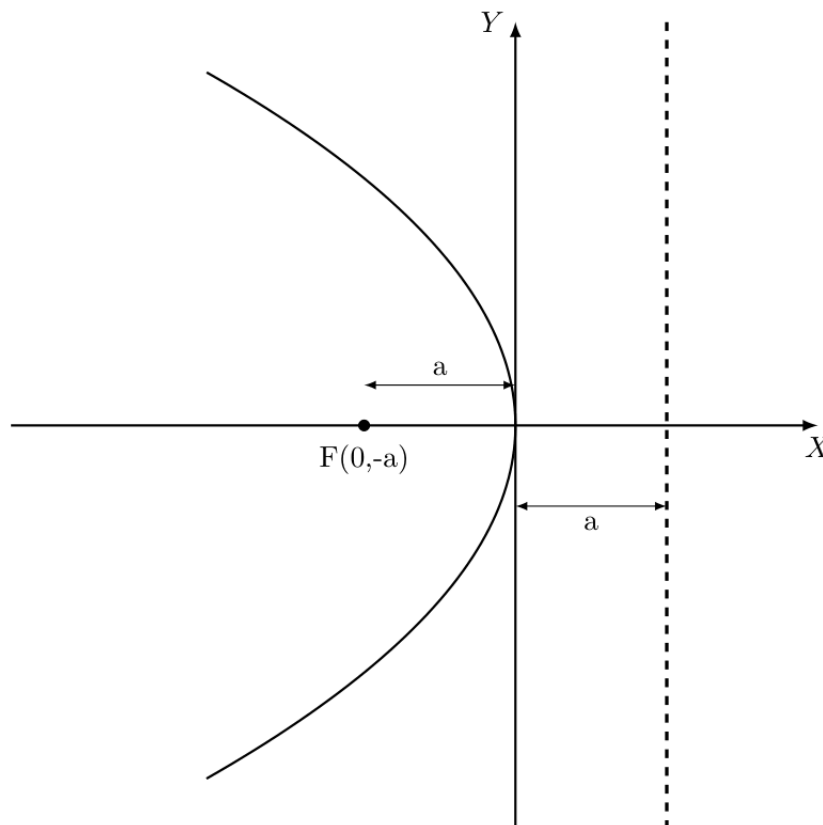


Figure 2:

3 Kepler's Laws

The scientist Kepler proposed three laws by observing the motion of planets around the sun. These laws are true for any central and r^{-2} type force. Suppose two objects of mass M and m are orbiting each other under the influence of gravitational force. [The three laws discussed in this section will only be true if $M \gg m$, because in this case the object of mass M remains almost stationary.]

- **First Law:** All planets move in elliptical orbits with the sun at one of the foci.

In the previous section, we derived orbits in a gravitational field and saw that all bound orbits are ellipses (a circle is also an ellipse where both axes are equal) and the source of the force was at the focus of the orbit.

- **Second Law:** A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time.

The angular momentum of the planet with respect to the sun is

$$l = mr^2 \frac{d\theta}{dt}$$

$$\Rightarrow \frac{l}{m} dt = r^2 d\theta$$

Now, the area swept during an angular displacement $d\theta$ is $dA = r dr d\theta = \frac{r^2}{2} d\theta$. Then,

$$\frac{l}{2m} dt = dA$$

Here, both l and m are constants. That is, the swept area dA is proportional to the elapsed time dt .

- **Third Law:** The square of the orbital period (T) of a planet is directly proportional to the cube of the semi-major axis (a) of its orbit. That is, $T^2 \propto a^3$.

If you think about this for a moment, it might initially seem like this law is incorrect, because it doesn't depend on the e of the orbit. Many orbits can exist with the same major axis a ; according to this law, the time periods of all those orbits are equal! This means, to know the period of a planet, you don't need to know its entire orbit, you just need to know the closest and farthest points from the sun to calculate the period. If you solve Exercise 2, you will see that the angular momentum of any elliptical orbit is-

$$l = m\sqrt{GMa(1 - e^2)}$$

From Kepler's second law we have seen

$$dA = \frac{l}{2m} dt$$

Substituting l here and integrating both sides-

$$\int dA = \frac{m\sqrt{GMa(1 - e^2)}}{2m} \int dt$$

Now, if the semi-major axis of an ellipse is a and its eccentricity is e , its area is $A = \pi ab = \pi a^2 \sqrt{(1 - e^2)}$. Therefore,

$$\pi a^2 \sqrt{(1 - e^2)} = \frac{\sqrt{GMa(1 - e^2)}}{2} T$$

$$\Rightarrow T = \frac{2\pi}{\sqrt{GM}} a^{\frac{3}{2}} \quad (6)$$

which proves Kepler's third law. In most cases, instead of using $T^2 \propto a^3$ for Kepler's third law, equation (6) is directly used.

4 General 2-Body Problem

In this section, we will determine the trajectories of two objects of masses m_1 and m_2 under the influence of gravitational force between them. m_1 and m_2 are comparable to each other, so both objects will remain in motion. The net force on this system of two masses is zero, so the combined center of mass of these two objects remains stationary. We will consider this stationary center of mass as the center of our coordinates and, as before, work in polar coordinates. Suppose, with respect to this coordinate system, the position vectors of m_1 and m_2 are $\vec{r}_1$ and $\vec{r}_2$ respectively. Then, according to the definition of the center of mass,

$$m_1\vec{r}_1 + m_2\vec{r}_2 = \vec{0} \quad (7)$$

Again, if we denote the separation vector between the two objects by $\vec{R}$, then

$$\vec{r}_2 - \vec{r}_1 = \vec{R} \quad (8)$$

By solving equations (7) and (8), $\vec{r}_1$ and $\vec{r}_2$ can be expressed in terms of $m_1, m_2, \vec{R}$. Solving them yields:

$$\vec{r}_1 = -\frac{m_2\vec{R}}{m_1 + m_2} \quad (9)$$

$$\vec{r}_2 = \frac{m_1\vec{R}}{m_1 + m_2} \quad (10)$$

Now, applying Newton's second law along the radial direction for both objects in this coordinate system, you will get-

$$m_1\ddot{\vec{r}}_1 = \frac{Gm_1m_2}{(\vec{r}_2 - \vec{r}_1)^3}(\vec{r}_2 - \vec{r}_1)$$

$$m_2\ddot{\vec{r}}_2 = -\frac{Gm_1m_2}{(\vec{r}_2 - \vec{r}_1)^3}(\vec{r}_2 - \vec{r}_1)$$

Using equation (8) and the second derivatives with respect to time of equations (9) and (10), we can write these two equations solely in terms of m_1, m_2 and $\vec{R}$. Substituting the values (be sure to calculate it yourself once!), an equation is obtained:

$$\frac{m_1m_2\ddot{\vec{R}}}{m_1 + m_2} = -\frac{Gm_1m_2}{\vec{R}^2}\hat{R}$$

We call $\frac{m_1m_2}{m_1+m_2}$ the reduced mass, and it is denoted by μ . And, if the total mass $m_1 + m_2$ is denoted by M , it can be written as-

$$\mu\ddot{\vec{R}} = -\frac{GM\mu}{\vec{R}^2}\hat{R} \quad (11)$$

Many of you may have understood by now the reason for doing this. We have managed to reduce the two equations of motion for the two bodies into just one variable $\vec{R}$! At first, it

might seem like we have lost some information in this process, but that hasn't happened. We essentially transformed the equations from r_1 and r_2 to R and R_{cm} . Because the value of R_{cm} and all its derivatives are already known, in the end, the equations became functions of R alone. If you look at equation (11), you will understand that this is the equation of an object of mass μ orbiting a stationary object of mass M ! This means that two objects of masses m_1 and m_2 orbiting each other is analogous to an object of mass μ orbiting a stationary object of mass M ! Note that the separation vector between M and μ here will be $\vec{R}$. Now, we can apply all the ideas we learned earlier for an object orbiting a stationary object here. Once the value of $\vec{R}$ is found, the position of each object relative to the center of mass can also be determined using equations (9) and (10).

Summary: Two Body Problem can be best solved with respect to the center of mass frame. In this frame, the equation for m_1 and m_2 orbiting each other is identical to the equation for a mass of $\frac{m_1 m_2}{m_1 + m_2}$ orbiting a stationary mass $(m_1 + m_2)$ at a distance of $(\vec{r}_2 - \vec{r}_1)$. Thus, instead of solving a two-body problem, solving a single-body problem is sufficient.

We can find $R(\theta)$ using the methods learned in the previous sections. However, remember that in the actual orbit, R is the distance between the two bodies, so $R(\theta)$ does not actually represent the orbit of either body. But using this, it is possible to find $r_1(\theta)$ and $r_2(\theta)$ from equations (9) and (10), which are the orbits of m_1 and m_2 respectively. In the reduced problem, we can also determine the orbital period using Kepler's third law. This will be the period of the orbit for both bodies, since their angular velocities are the same. You can easily show yourselves that if the angular velocities of the two objects are not identical, the center of mass would not remain stationary.

Example: Two stars of masses M_1 and M_2 are rotating in circular orbits around their combined center of mass. Their velocities with respect to the center of mass of the system are v_1 and v_2 respectively. Determine the orbital period of the stars.

Solution: Since there is no condition like $M_1 \gg M_2$ or $M_1 \ll M_2$, this is essentially a two-body problem. We will first convert this into a single-body problem. Their reduced mass,

$$\mu = \frac{M_1 M_2}{M_1 + M_2}$$

Therefore, we can say that a star of mass μ is orbiting a stationary star of mass $M_1 + M_2$, and let the distance between them be R . In that case, according to Kepler's third law,

$$T = \frac{2\pi}{\sqrt{2G(M_1 + M_2)}} R^{\frac{3}{2}} \quad (12)$$

Now, our task is to express R in terms of v_1 and v_2 . R is the total distance between the stars in the original problem.

In the figure, point O is the combined center of mass of the system. For equation (7) to hold true, $\vec{r}_1$ and $\vec{r}_2$ must always lie on the same line. Therefore, the line $M_1 O M_2$, the angular velocities of M_1 and M_2 will be the same. For this reason, the time periods of both stars will also be identical. Let $\overline{OM_1} = \vec{r}_1$, $\overline{OM_2} = \vec{r}_2$ and $\overline{M_1 O M_2} = \vec{r}_2 - \vec{r}_1 = \vec{R}$. Applying Newton's second law along the radial direction with respect to center O,

$$\frac{M_1 v_1^2}{|\vec{r}_1|} = \frac{GM_1 M_2}{\vec{R}^2}$$

Using the equations for the center of mass and $\vec{R}$, similar to equation (9), we can write,

$$\vec{r}_1 = -\frac{M_2 \vec{R}}{M_1 + M_2}$$

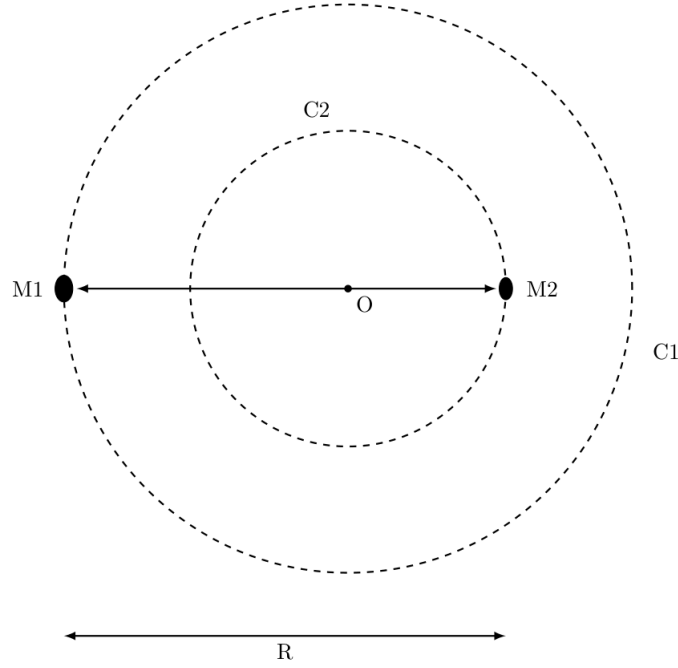


Figure 3: Binary star system in circular orbit

Substituting this into the previous equation,

$$\frac{M_1(M_1 + M_2)v_1^2}{M_2|\vec{R}|} = \frac{GM_1M_2}{\vec{R}^2}$$

$$\Rightarrow |\vec{R}| = \frac{GM_2^2}{(M_1 + M_2)v_1^2}$$

Substituting this value into equation (12),

$$T = \frac{2\pi}{\sqrt{2G(M_1 + M_2)}} \frac{G^{\frac{3}{2}}M_2^3}{(M_1 + M_2)^{\frac{3}{2}}v_1^3} = \frac{\sqrt{2}\pi GM_2^3}{(M_1 + M_2)^2v_1^3}$$

Note that if we had applied Newton's second law on M_2 , we would have gotten the same orbital period, but it would be a function of v_2 . This problem could actually be solved by knowing only v_1 or v_2 .

5 Conservation of Energy and Angular Momentum

Conserved quantities play an important role in solving any type of problem. It has already been discussed that angular momentum is conserved for central forces. Since a two-body problem can also be reduced to a single-body problem, we will only discuss the single-body case, where M is stationary and mass m is orbiting M . In this case, the total energy

$$E = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - \frac{GMm}{r}$$

For any r , $\dot{r}$, and $\dot{\theta}$ at any given moment, the value of E is the same. You can show (Exercise 2) that the total energy in any elliptical orbit is $\frac{-GMm}{2a}$. When applying conservation of angular momentum, you also must note if any torque is coming from another source. For example,

suppose two satellites are orbiting the Earth in the same orbit. If they collide with each other, one will exert a torque on the other (here our coordinate center is the Earth). However, if the two satellites are considered together as one system, the total angular momentum of that system will be conserved, since the sum of the torques acting on both satellites is zero. The $\dot{\theta}$ term in the total energy can be substituted using angular momentum.

$$l = m\dot{\theta}r^2$$

Then, total energy

$$E = \frac{1}{2}m\dot{r}^2 + \frac{l^2}{2mr^2} - \frac{GMm}{r}$$

The energy equation for this potential $\frac{l^2}{2mr^2} - \frac{GMm}{r}$ is exactly like an object moving in one dimension along r . This potential is called the effective potential,

$$U_{eff} = \frac{l^2}{2mr^2} - \frac{GMm}{r}$$

Through this, we have essentially reduced two-dimensional motion down to one dimension! This also shows that to know the orbit of any object under gravitational force, it is sufficient to know only its total energy, angular momentum, and the mass of the object. This is why if energy and angular momentum are known or can be determined beforehand in a problem, one should check if the problem can be solved using their conservation. For example, a very common type of question is finding the velocity or distance at one point in an orbit given the velocity, position, etc. at another point. Such problems can generally be solved using just the conservation of energy and angular momentum! Apart from these, in solving any problem, one should write down the conserved quantities right at the beginning; in most cases, this makes the calculations much easier.

Example: [BDPhO Camp Exam 2021]¹ An object of mass m is in a circular orbit of radius R under the influence of force $\vec{F} = -GMmr^{-n}\vec{\hat{r}}$ ($M \gg m$). The object in orbit is given a small nudge along its radius. If $n = 2$, as a result, the object of mass m will execute simple harmonic motion while revolving in its circular path. What is the period of this simple harmonic motion? For what values of n will the motion of m be stable upon a small radial nudge?

Solution: Since we are only concerned with motion along the radius, we will substitute out the angular motion. As a result, we will get the effective potential seen earlier. In the Simple Harmonic Motion chapter, we will see that if an object of mass m undergoes simple harmonic motion due to a potential $U(r)$, its angular frequency is

$$\omega_0 = \sqrt{\frac{U''(r_0)}{m}}$$

where U'' denotes the second derivative of the potential with respect to r , or $\frac{d^2U}{dr^2}$. And, r_0 is the equilibrium point of the object, which is usually found by solving the equation $\frac{dU}{dr} = 0$ (details in the Oscillation chapter). To solve the first part, we have already seen,

$$U_{eff} = \frac{l^2}{2mr^2} - \frac{GMm}{r}$$

¹In this problem we will use some properties of simple harmonic motion. Before solving the problem, read the Oscillation chapter ().

Again, since the radius of the orbit is R ,

$$\begin{aligned}\frac{d}{dr}U_{eff}\Big|_{r=R} &= 0 \\ \Rightarrow -\frac{l^2}{mR^3} + \frac{GMm}{R^2} &= 0 \\ \Rightarrow l^2 &= GMm^2R\end{aligned}$$

Substituting this equation for angular momentum into the effective potential equation,

$$U_{eff} = \frac{GMm^2R}{2mr^2} - \frac{GMm}{r} = GMm\left(\frac{R}{2r^2} - \frac{1}{r}\right)$$

Then we can now also determine the second derivative of the effective potential:

$$U'' = GMm\left(\frac{3R}{r^4} - \frac{2}{r^3}\right)$$

Substituting $r = R$ into this gives $U''(R) = \frac{GMm}{R^3}$. Since $T = \frac{2\pi}{\omega_0}$,

$$T = 2\pi\sqrt{\frac{R^3}{GM}}$$

An interesting thing in this solution is that the period of motion along this radius is exactly equal to the period of its angular motion (orbital period)! By drawing the initial and final orbits of the object in the same figure, you will see that the final orbit is also essentially a circle, but the center of this circle is slightly shifted from the center of the initial orbit.

We can solve the second part of this question in the same way. For this, we need to know another concept from the Oscillation chapter; that is, if an equilibrium point of an object is R , then if $U''(R) > 0$, the equilibrium point is stable (since the potential energy is minimum), and if $U''(R) < 0$, the equilibrium point is unstable (since the potential energy is maximum). In this case, it is given that $F = -GMmr^{-n}$. The potential for this force is,

$$U_f = \int_{\infty}^r F dr = -\frac{GMmr^{-n+1}}{n-1}$$

Then the effective potential is, $U_{eff} = \frac{l^2}{2mr^2} - \frac{GMmr^{-n+1}}{n-1}$. If the radius of the circular orbit is R ,

$$\begin{aligned}\frac{d}{dr}U_{eff}\Big|_{r=R} &= 0 \\ \Rightarrow -\frac{l^2}{mR^3} + \frac{GMm}{R^n} &= 0 \\ \Rightarrow l^2 &= \frac{GMm^2}{R^{n-3}}\end{aligned}$$

Again, by differentiating the equation of the effective potential twice and substituting l^2 ,

$$U''(R) = \frac{(3-n)GMm}{R^{n+1}}$$

From here you can see that $U''(R) > 0$ only when $n < 3$. Therefore, the condition for the orbit to be stable must be $n < 3$.

6 Exercises

6.1 Eccentricity Equation

The orbit of an object under the influence of gravitational force is given by equation (4) and its total energy is E and total angular momentum is l . Determine the equation for e as a function of E, l, M , and m . (Hint: Since the radial velocity is zero, the $\theta = 0$ point of the orbit is very convenient for applying the conservation of energy.)

6.2 Conserved quantities in elliptical orbit

An object of mass m is revolving in an elliptical orbit centered around an object of mass M . Determine the total energy and angular momentum of mass m as a function of the ellipse parameters and the masses of the two objects. Assume $M \gg m$.

6.3 Binary Star system

The masses of the two stars in a binary star system are m_1 and m_2 respectively. If their total energy in their center of mass frame is E , determine the orbital period T of the stars.

6.4 Stable Orbit Condition

Is it possible to make an object of mass m revolve in a circular orbit of radius r under the influence of a force of the form $\vec{F} = -\alpha r^{-n} \hat{r}$? If possible, determine the necessary conditions for it. Here α is a positive constant and n can be any real number.